

3577A PROGRAMMING CODES

APPENDIX D

DISPLAY FORMAT

Function	HP-IB code
TRACE 1	TR1
TRACE 2	TR2
DISPLAY FUNCTION	DSF *
Log Magnitude	DF7
Linear Magnitude	DF6
Phase	DF5
Polar	DF4
Real	DF3
Imaginary	DF2
Delay	DF1
Trace Off	DF0
Delay Aperture menu	DAP *
Aperture .5% of span	AP1
Aperture 1% of span	AP2
Aperture 2% of span	AP3
Aperture 4% of span	AP4
Aperture 8% of span	AP5
Aperture 16% of span	AP6
Return	RET *
INPUT	INP *
Input = R	INR
Input = A	INA
Input = B	INB
Input = A/R	IAR
Input = B/R	IBR
Input = D1	ID1
Input = D2	ID2
Input = D3	ID3
Input = D4	ID4
Return	RET *
User Defined Input	UDI
Input = S_{11}	I11
Input = S_{21}	I21
Input = S_{12}	I12
Input = S_{22}	I22
Copy Input	CPI
Test Set Forward	TSF
Test Set Reverse	TSR
SCALE	SCL *
Autoscale	ASL
Reference Level (entry)	REF
Scale /DIV (entry)	DIV
Reference Position (entry)	RPS
Reference Line Off	RL0
Reference Line On	RL1
Copy Scale	CPS

Phase Slope (entry)
Phase Slope Off
Phase Slope On
Polar Full Scale (entry)
Polar Phase Ref (entry)
Smith Chart Off
Smith Chart On

MARKER

Marker Position (entry)
Marker Off
Marker On
Zero Marker
Marker Offset Off
Marker Offset On
Marker Offset (entry)
Marker Offset Freq (entry)
Marker Offset Amp (entry)
Marker Coupling Off
Marker Coupling On
Polar Mag Offset (entry)
Polar Phase Offset (entry)
Polar Real Offset (entry)
Polar Imag Offset (entry)
Polar Marker Units (Re/Im)
Polar Marker Units (Mg/Ph)

MARKER →

MKR → Reference Level
MKR → Start Frequency
MKR → Stop Frequency
MKR → Center Frequency
MKR Offset → Span
MKR → Max
MKR → Min
MARKER SEARCH menu
MKR Target Value (entry)
MKR → Right for Target
MKR → Left for Target
Return
MKR → Full Scale
MKR → Polar Phase Ref

STORE DATA

Store in register D1
Store in register D2
Store in register D3
Store in register D4
Store and Display
User defined store
Store to D1
Store to D2
Store to D3
Store to D4

PSL
PS0
PS1
PFS
PPR
GT0
GT1

MKR *
MKP
MR0
MR1
ZMK
MO0
MO1
MKO
MOF
MOA
CO0
CO1
PMO
PPO
PRO
PIO
MRI
MMP

MKG *
MTR
MTA
MTB
MTC
MOS
MTX
MTN
MSM *
MTV
MRT
MLT
RET *
MTP
MPF

STO *
SD1
SD2
SD3
SD4
STD
UDS
TD1
TD2
TD3
TD4

* Use not required. The only function of this code is to display a menu (if bus diagnostics are on).

MEASUREMENT CALIBRATION

Normalize
 Normalize (Short)
 Calibrate, Partial
 Calibrate, Full
 Continue Calibration

DEFINE MATH

Constant K1, Real
 Constant K1, Imaginary
 Constant K2, Real
 Constant K2, Imaginary
 Constant K3, Real
 Constant K3, Imaginary
 Define Function
 Function F1
 Function F2
 Function F3
 Function F4
 Function F5
 Math term for input R
 Math term for input A
 Math term for input B
 Math term for storage reg
 Math term for constant
 Math term for function
 Math bracket
 Math function plus
 Math function minus
 Math function multiply
 Math function divide
 Math bracket
 Return

DATA ENTRY SECTION COMMANDS

Increment (up arrow)
 Decrement (down arrow)
 Continuous Entry (knob) Off
 Continuous Entry (knob) On
 Entry Off

DISPLAY FORMAT SUFFIX UNITS

dBm
 dBV (rms)
 dB relative
 Volt (rms)
 milli-Volt (rms)
 micro-Volt (rms)
 nano-Volt (rms)
 degrees
 degrees/span
 radians
 radians/span
 seconds
 milliseconds
 microseconds
 nanoseconds
 percent
 degrees/span
 radians/span
 MHz
 kHz
 Hz
 exponent

CAL *

NRM
 NRS
 CPR
 CFL
 CGO

DFN *

KR1
 KI1
 KR2
 KI2
 KR3
 KI3

DFC *

UF1
 UF2
 UF3
 UF4
 UF5

R

A

B

D

K

F

(

+

-

*

/

)

RET*

IUP

IDN

CE0

CE1

HLD

DBM

DBV

DBR

V

MV

UV

NV

DEG

DSP

RAD

RSP

SEC

MSC

USC

NSC

%

DSP

RAP

MHZ

KHZ

HZ

E

SOURCE**Function****HP-IB code****SWEEP TYPE**

Linear Sweep
 Alternate Sweep
 Log Sweep
 Amplitude Sweep
 CW
 Sweep Direction Up
 Sweep Direction Down

STY *

ST1

ST2

ST3

ST4

ST5

SUP

SDN

SWEEP MODE

Continuous
 Single Sweep
 Manual Sweep
 Manual Frequency (entry)
 Manual Amplitude (entry)
 Marker → Manual

SMD *

SM1

SM2

SM3

MFR

MAM

MTM

SWEEP TIME

Sweep Time (entry)
 Step Time (entry)
 Sample Time (entry)

STM *

SWT

SMT

MSR

FREQUENCY

Source Frequency (entry)
 Start Frequency (entry)
 Stop Frequency (entry)
 Center Frequency (entry)
 Frequency Span (entry)
 FRC Step size (entry)
 Sweep Resolution menu
 Freq Swp Res 51 pts/span
 Freq Swp Res 101 pts/span
 Freq Swp Res 201 pts/span
 Freq Swp Res 401 pts/span
 Return
 Full Sweep
 Freq Step Size (entry)

FRQ *

SFR

FRA

FRB

FRC

FRS

CFS

SRL *

RS1

RS2

RS3

RS4

RET *

FSW

FST

AMPLITUDE

Source Amplitude (entry)
 Amp Step Size (entry)
 Clear Trip, Source
 Start Amplitude (entry)
 Stop Amplitude (entry)
 Steps/Sweep menu
 Number of steps = 6
 Number of steps = 11
 Number of steps = 21
 Number of steps = 51
 Number of steps = 101
 Number of steps = 201
 Number of steps = 401
 Return
 Full Sweep

AMP *

SAM

AST

CTS

AMA

AMB

NST *

NS1

NS2

NS3

NS4

NS5

NS6

NS7

RET *

FSW

* Use not required. The only function of this code is to display a menu (if bus diagnostics are on).

TRIGGER MODE

Free Run
Line Trigger
External Trigger
Immediate

SWEEP TRIGGER**SWEEP RESET**

} TRG/
RESET

SOURCE SUFFIX UNITS

dBm
dBV (rms)
Volt (rms)
milli-Volt (rms)
micro-Volt (rms)
nano-Volt (rms)
seconds
milliseconds
MHz
kHz
Hz
exponent

TRM *

TG1
TG2
TG3
TG4

TRG**RST**

DBM
DBV
V
MV
UV
NV
SEC
MSC
MHZ
KHZ
HZ
E

Length A (entry)
Length A Off
Length A On
Length B (entry)
Length B Off
Length B On
Length Step Size (entry)

LNA

LA0
LA1
LNB
LB0
LB1
LNS

RECEIVER SUFFIX UNITS

meters
centimeters
seconds
milliseconds
microseconds
nanoseconds
exponent

MET
CM
SEC
MSC
USC
NSC
E

INSTRUMENT STATE**RECEIVER****Function****HP-IB code****RESOLUTION BW**

Resolution BW 1 Hz
Resolution BW 10 Hz
Resolution BW 100 Hz
Resolution BW 1 kHz
Auto Bandwidth Off
Auto Bandwidth On

RBW *
BW1
BW2
BW3
BW4
AU0
AU1

AVERAGE

Averaging Off
N=4
N=8
N=16
N=32
N=64
N=128
N=256

AVE *
AV0
AV1
AV2
AV3
AV4
AV5
AV6
AV7

ATTENUATION

Attenuation R = 0 dB
Attenuation R = 20 dB
Attenuation A = 0 dB
Attenuation A = 20 dB
Attenuation B = 0 dB
Attenuation B = 20 dB
Impedance R = 50 Ω
Impedance R = 1 M Ω
Impedance A = 50 Ω
Impedance A = 1 M Ω
Impedance B = 50 Ω
Impedance B = 1 M Ω
Clear Trip, Receiver

ATT *
AR1
AR2
AA1
AA2
AB1
AB2
IR1
IR2
IA1
IA2
IB1
IB2
CTR

LENGTH

Length R (entry)
Length R Off
Length R On

LEN *
LNR
LR0
LR1

Function**HP-IB Code****SPECIAL FUNCTIONS**

Confid. (self) test menu
Self test channel R
Self test channel A
Self test channel B
Return
Beeper off
Beeper on
Service Diagnostics menu
Source Leveling off
Source Leveling on
Settling Time off
Settling time on
Synthesizer Diag off
Synthesizer Diag on
Display Test Pattern
Trace Memory Test
Fast Processor Test
I/O port test
More Serv Diag menu
Display Memory Test
Software Revision message
Return
S-Parameters Off
S-Parameters On

SPC *
SLF *
STR
STA
STB
RET *
BP0
BP1
SDG *
SL0
SL1
SE0
SE1
SY0
SY1
DTP
TMT
FPT
PRT
MOR *
DST
SRV
RET *
SP0
SP1

SAVE INSTRUMENT STATE

Save state in register 1
Save state in register 2
Save state in register 3
Save state in register 4
Save state in register 5

SAV *
SV1
SV2
SV3
SV4
SV5

RECALL INSTRUMENT STATE

Recall old (last) state
Recall register 1
Recall register 2
Recall register 3
Recall register 4
Recall register 5

RCL *
RLS
RC1
RC2
RC3
RC4
RC5

* Use not required. This code's only function is to display a menu (if but diagnostics are on).

INSTRUMENT PRESET**PLOT MENU**

Plot all
 Plot trace 1
 Plot trace 2
 Plot graticule
 Plot characters
 Plot trace 1 marker
 Plot trace 2 marker
 Configure Plot menu
 Trace 1 linetype (entry)
 Trace 2 linetype (entry)
 Trace 1 pen number (entry)
 Trace 2 pen number (entry)
 Graticule pen no. (entry)
 Pen speed fast (max)
 Pen speed slow
 Set plot config to default
 Return

IPR**PLM ***

PLA
PL1
PL2
PLG
PLC
PM1
PM2
CPT *
T1L
T2L
T1P
T2P
PGP
PNM
PNS
PLD
RET *

Dump average number
 Dump key or knob
 Dump characters
 Dump Instrument ID

 Load register A
 Load register B
 Load register R
 Load register D1
 Load register D2
 Load register D3
 Load register D4
 Load state (learn mode in)

 Graticule off
 Graticule on
 Characters off
 Characters on
 Annotation off
 Annotation on
 Annotation Clear
 Menu off
 Menu on
 Menu clear

DAN
DKY
DCH
ID?

LRA
LRB
LRR
LD1
LD2
LD3
LD4
LMI

GR0
GR1
CH0
CH1
AN0
AN1
ANC
MN0
MN1
MNC

HP-IB ONLY COMMANDS**Function****HP-IB code**

Settling Time Entry

 Dump register A
 Dump register B
 Dump register R
 Dump register D1
 Dump register D2
 Dump register D3
 Dump register D4
 Dump trace 1
 Dump trace 2
 Dump marker 1
 Dump marker 2
 Dump marker 1 position
 Dump marker 2 position
 Dump state (learn mode out)
 Dump status

STE

DRA
DRB
DRR
DD1
DD2
DD3
DD4
DT1
DT2
DM1
DM2
MP1
MP2
LMO
DMS

ASCII data format
 64 bit IEEE data format
 32 bit HP 3577A binary
 Bus diagnostics mode off
 Bus diagnostics on, fast
 Bus diagnostics on, slow

 Enter Menu (user defined)
 Enter Annotation
 Enter Graphics
 Clear Keyboard Buffer
 Take Measurement
 Set SRQ Mask
 Error Reporting mode 0
 Error Reporting mode 1
 Error Reporting mode 2
 Error Reporting mode 3
 Send SRQ

FM1
FM2
FM3
BD0
BD1
BD2

ENM
ENA
ENG
CKB
TKM
SQM
ER0
ER1
ER2
ER3
SRQ

* Use not required. The only function of this code is to display a menu (if bus diagnostics are on).